

When transporters own the road, how much can the network carry in a day?

The road-occupancy model and the network bound: thesis chapter 2 and the bound section of chapter 3

A capacity bound for shipyard road networks with shareable widths and asymmetric loaded and empty occupancy

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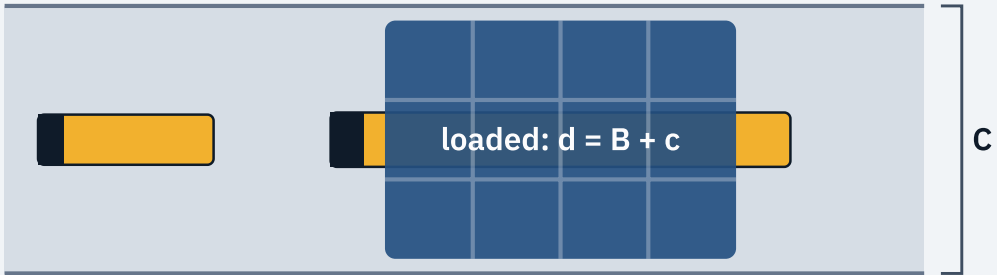
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System T1' and the route-free T1"; none of over 50,000 occupancy simulation points in two yards exceeds the bound · thesis-core a882742

One resource = one width container

At every moment: widths in use $\leq$ capacity

capacity $C = \text{road width } W + \text{lateral clearance } c$



empty: not enough width left, waits

This resource over one day ($H = 16 \text{ h}$)



$$\sum \ell \cdot u(d / C) \leq H$$

One row per resource: this is the LP of the network bound

This sub-study in one minute

1

The question

When loaded transporters hold segments and junctions exclusively, how many tasks a day can a yard's network complete? The bound should hold for a whole class of traffic rules and show where it binds.

2

The model

Segments, junctions and stops are width containers, held for a blocking time (passage plus clearance, plus the turn sweep at junctions); loaded and empty vehicles take different widths.

3

The bound T1'

The occupancy-rate LP plus width bin-packing rows, crane rows and a window edge term. System T1', free of sampling noise: Okpo 1,736, Yantai 1,298 a day (main scenario); none of over 50,000 simulation points exceeds it.

4

Route-free T1''

Loaded routes are chosen by the LP too, so the bound holds for any routing rule, with a certificate that does not rest on solver tolerances. Route slack in the bound runs from 0 (an entrance binds) to +45% (a through corridor).

The bound is the upper edge of the capacity interval and names where capacity binds; T27 now re-checks the lower edge.

The order of this briefing (click to jump)

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The question: write road occupancy as a structural capacity constraint and compute the daily throughput the network can carry

Place in the thesis

Chapter 2: the occupancy model. Chapter 3's bound section: T1' and its checks, the system T1', the route-free T1'' and route slack.

C1 Two bounds and the capacity interval

Throughput $\leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{ network bound}\}$. The network bound is set by the blocking-time structure and can be computed by linear programming. Attainable capacity lies within the interval [best deadlock-free rule, network bound].

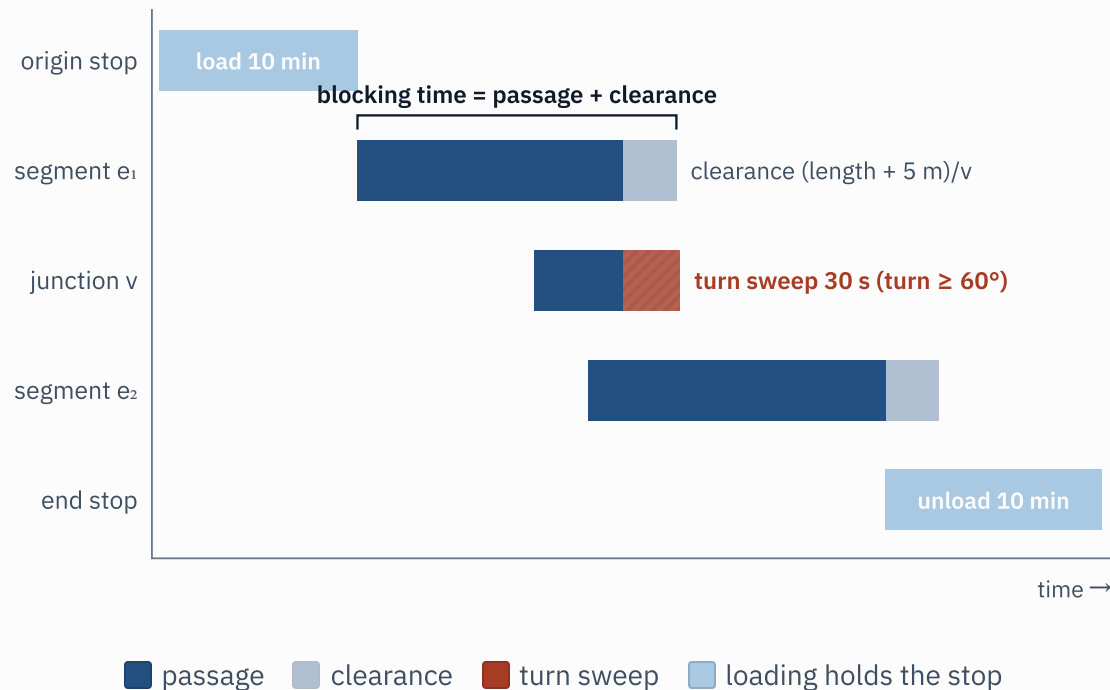
Also feeds

C4, the exchange rate: dual prices are its input, and plateau over bound separates C3 from C4. C5 and H6, cross-yard prediction: the bound computed from the map, with its binding elements, is the predictor.

ITEM	STATEMENT (MANIFESTO V1, §4)	STATUS
T1'	Network part of the two bounds: occupancy-rate LP with width bin-packing rows, crane rows and the window edge term. Holds for rules satisfying (A1)–(A3); loaded routes fixed	draft; author rewrites
T1''	Route-free bound: loaded moves also as multicommodity flow; holds for any routing rule	T23 draft; author
T5	Safe orchestration: reserving up to the next safe holding point cannot deadlock, with a loss bound relative to the upper bound	tested in T27
T6	Road–vehicle exchange rate: given by dual prices; upgrades pay only on tight constraints. Use T1'' for new links	author
Complexity	Feasibility with deadlines and exclusive resources is NP-hard (reduction from the blocking job shop)	to do
Candidate	Dock-mouth proposition: at nominal load $\rho < 1$, starvation caused by blocking destabilises the system; waiting off the road or using the dock road in sections restores stability. The other is attainability of T1'	author, after T22

Sources: manifesto v1 (tag manifesto-v1, docs/研究宣言.md), §2 C1 and the §4 theorem list, quoted in full; status as of 10 Oct 2026 (T1'' draft docs/T23_route_free_proof.md; T5 tested in T27; dock readings in the T22 report §3); chapter plan from the direction document (8 Oct 2026), §4.

Resources are width containers: segments, junctions and stops are held for a blocking time



- **Three kinds of resources:** segments (721 on Okpo), junctions and entrance nodes (552), drive-in stops (172); from T20 the cranes are dock resources too
- **Capacity** = road width W + lateral clearance c (1 m); a vehicle takes its outline width + c ; at every moment the widths in use stay within capacity
- **Blocking time:** a segment is held from the head entering until the tail clears the next junction, plus longitudinal clearance; a junction for $(D + \text{length} + 5 \text{ m})/v$, plus a 30 s sweep for turns $\geq 60^\circ$
- **Stops:** shops load at their doors (entrance nodes); stockyards and assembly areas have drive-in stops; buffers and docks load at the roadside, so that road segment is the stop

$$\sum_v d_{v,r}(t) \leq C_r \text{ for every resource } r \text{ and time } t$$

Loaded and empty occupancy are asymmetric: passability classes follow from widths, and widths and empty-run rules are scanned as scenarios



■ loaded: block width $B = 10$ m ■ empty: platform $w = 5.5$ m same scale; $c = 1$ m

- **Blocks** drawn with width $B \in \{5, \dots, 15\}$ m and length $\max(B, 10) - 30$ m; loaded outline width $\max(B, 5.5)$ m. Empty platform 16.25×5.5 m; 12 km/h empty, 6 km/h loaded
- **No precise survey:** road widths are graded from imagery (about ± 1.5 m); the three classes follow from widths, not from road-by-road judgement

SCENARIO	OKPO T1'	YANTAI T1'
Main scenario	1,720	1,252
Widths +1.5 m	1,789 +4%	1,252 ±0%
Widths -1.5 m	1,674 -3%	1,248 ±0%
Empties exclusive everywhere	1,720 ±0%	1,252 ±0%
Empties pass everywhere	1,822 +6%	1,252 ±0%

Yantai binds at door nodes, so neither widths nor the empty-run rule move it; Okpo binds on narrow roads and responds to both.

Sources: thesis-core a882742, docs/T17_flag_report.md §1.1, §1.3 and the §2 parameter table (block sizes as in MY-B; speeds from the Suzhou Dafang DCY270 data sheet); T1' in the table is the mean of 10 seeds in results/a0ab16fceb94/bounds.csv (Okpo) and b82aa19b22af/bounds.csv (Yantai), tasks a day. Sketch drawn to one scale.

Four traffic rules, and the conditions (A1)–(A3) under which the bound holds

Free flow	No capacity check, vehicles never interact; the field's default. The bound does not apply: on a one-lane example it is exceeded already at $K = 4$	Conditions (proof draft, §3) (A1) Each task holds each resource at least for its no-wait window and at least at its outline width (A2) Empty runs occupy resources at least as much as the LP's empty-run coefficients (A3) The windows counted for one vehicle never overlap: the end window stops when unloading ends (A1') For T1": (A1) on whatever route is actually taken; the occupancy rules govern requests and releases only, not how the route was chosen
Whole-route reservation (just-in-time departure)	Empty run, loading, loaded run and unloading booked at once; waits only at stops, never on the way; deadlock-free by construction	
Segment request · reference rule (teleport clearing)	At each junction request “junction + next segment”, waiting on the current segment if refused; a deadlock sends one vehicle off the network for 900 s. Reference only	
Safe release, segment_safe T22, in hand	Segment request with a Banker-type safety check: only requests that cannot lead to deadlock are granted, no teleport. 860,291 decisions identical to a naive reduction, never a deadlock; conservative: about 2/3 of its waits start from a criterion refusal, and it falls below reservation for $K \geq 20$	

Whole-route reservation

Segment request while no clearing occurs

Reference rule (teleport clearing): a teleport can cut a hold short, so (A1) may fail

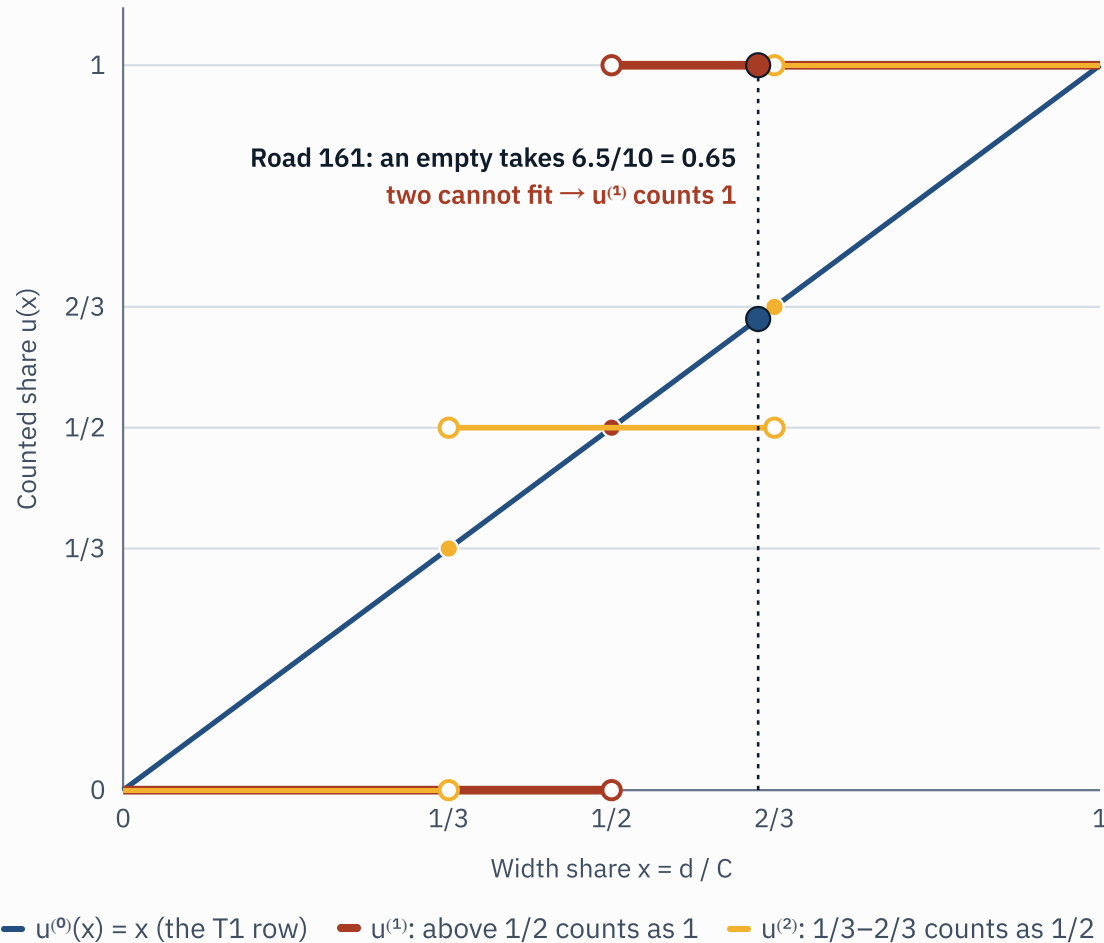
segment_safe: no teleport and never a deadlock, so segment request without clearing; waiting only lengthens holds

Free flow: no capacity check

All four share one dispatcher: nearest idle vehicle, first come first served. The fleet bound $K \cdot H/\bar{c}$ holds for all four. In both yards the best deadlock-free rule is whole-route reservation (T22); a less conservative deadlock-free rule is tested in T27.

Sources: thesis-core a882742, docs/T18_T1_tight_proof.md §1, §3, §5 (segment request without clearing is inside the proof); docs/T17_flag_report.md §1.2 and §0 (one-lane counterexample); docs/T22_report.md §2.1 (segment_safe: the three-step check, results/t22_report_3287d62c7916/t22_naive.json, the ring6 counterexample, the share of criterion refusals) and §4 (best deadlock-free rule); docs/T23_route_free_proof.md §1 (A1').

T1' : three new kinds of rows tighten the LP bound and bring in the dock



$$\begin{aligned}
 &\max \lambda \\
 &\text{s.t. } \lambda \cdot \bar{a}_r^{(k)} + \sum_a f_a \cdot e_{ar}^{(k)} \leq H \quad \forall r, k = 0, 1, 2 \\
 &\quad \sum_{a \rightarrow n} f_a - \sum_{n \rightarrow a} f_a = \lambda \cdot (o_n - d_n) \quad \forall n \\
 &\quad \lambda \cdot s_z \cdot 190 \min \leq m_z \cdot H \quad \forall \text{dock } z
 \end{aligned}$$

Width bin-packing rows

T18

The share $x = d/C$ becomes the Fekete–Schepers dual-feasible functions $u^{(1)}, u^{(2)}$, in exact rationals: two vehicles that cannot fit side by side each count as exclusive

Crane rows

T20

One row per dock: an erection holds a crane 190 min (hook 10, position and release 180) against cranes $\times 16$ h

Window edge term

T21

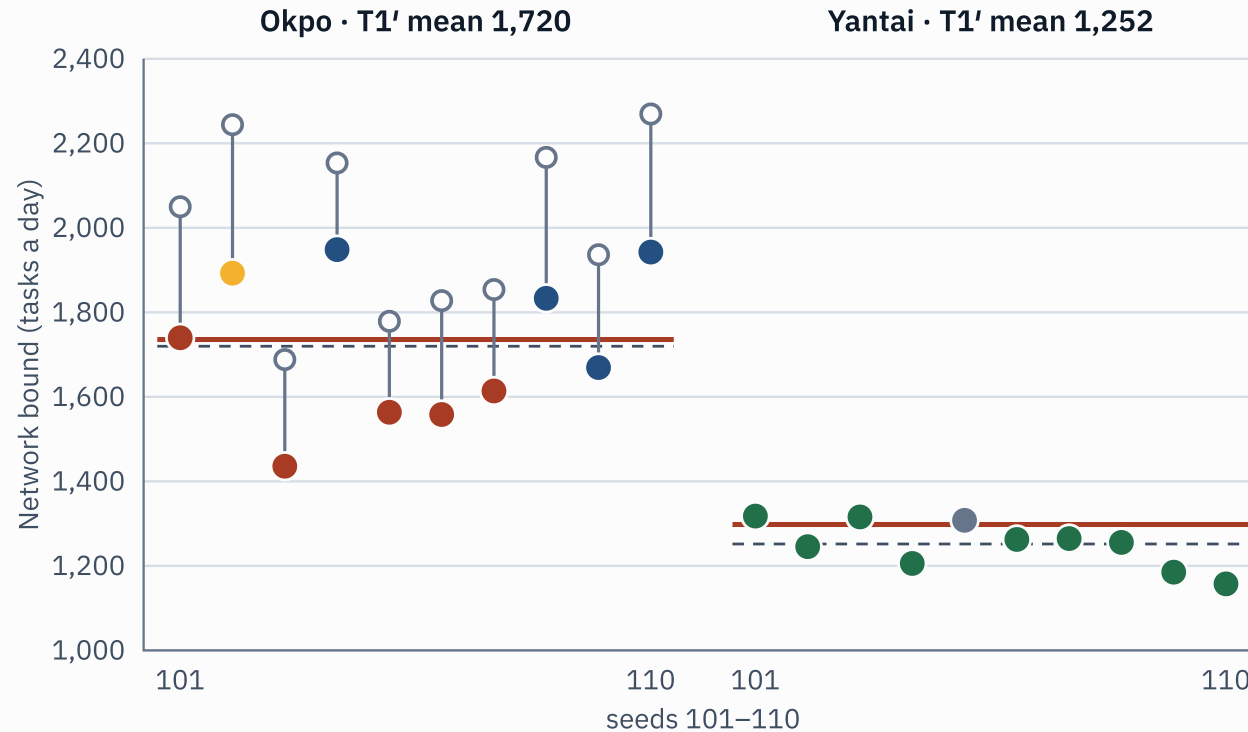
The case check runs over a finite window W , where a crane can start once more than its rate: each lift counts $190 \min \times W / (W + 190 \min)$

Theorem T1': for a rule satisfying (A1)–(A3), long-run daily throughput $\lambda \leq \min\{T1', K \cdot H / \bar{c}\}$; loaded routes as in the simulation (shortest paths).

Sources: thesis-core a882742, docs/T18_T1_tight_proof.md §1–3 (lemma 1 is a special case of Fekete & Schepers 2001, proved on its own in the draft); crane rows in the T20 report §0 item 3, the 190 min from Zhang 2025, lines 2525–2528 (another yard); the window edge term in the T21 report §5. $\bar{a}$ and e are the no-wait holds of one task and one empty run on resource r (seconds $\times$ share u); s_z is the erection share to dock z and m_z its cranes.

The chart is drawn from the definitions.

Seed by seed in two yards: Okpo tightens by 14% and binds on roads; Yantai binds at the two doors of one building



- T1 (hollow)
- T1' · road 161
- T1' · road 010-6
- T1' · road 026
- T1' · building 014 doors
- T1' · building 007 door
- mean of 10 seeds (dashed)
- system T1' (solid, T23)

1,997 → 1,720

Okpo main scenario, T1 → T1', mean of 10 seeds (a day, -14%); seeds range 1,436-1,948, system T1' 1,736

802 → 703

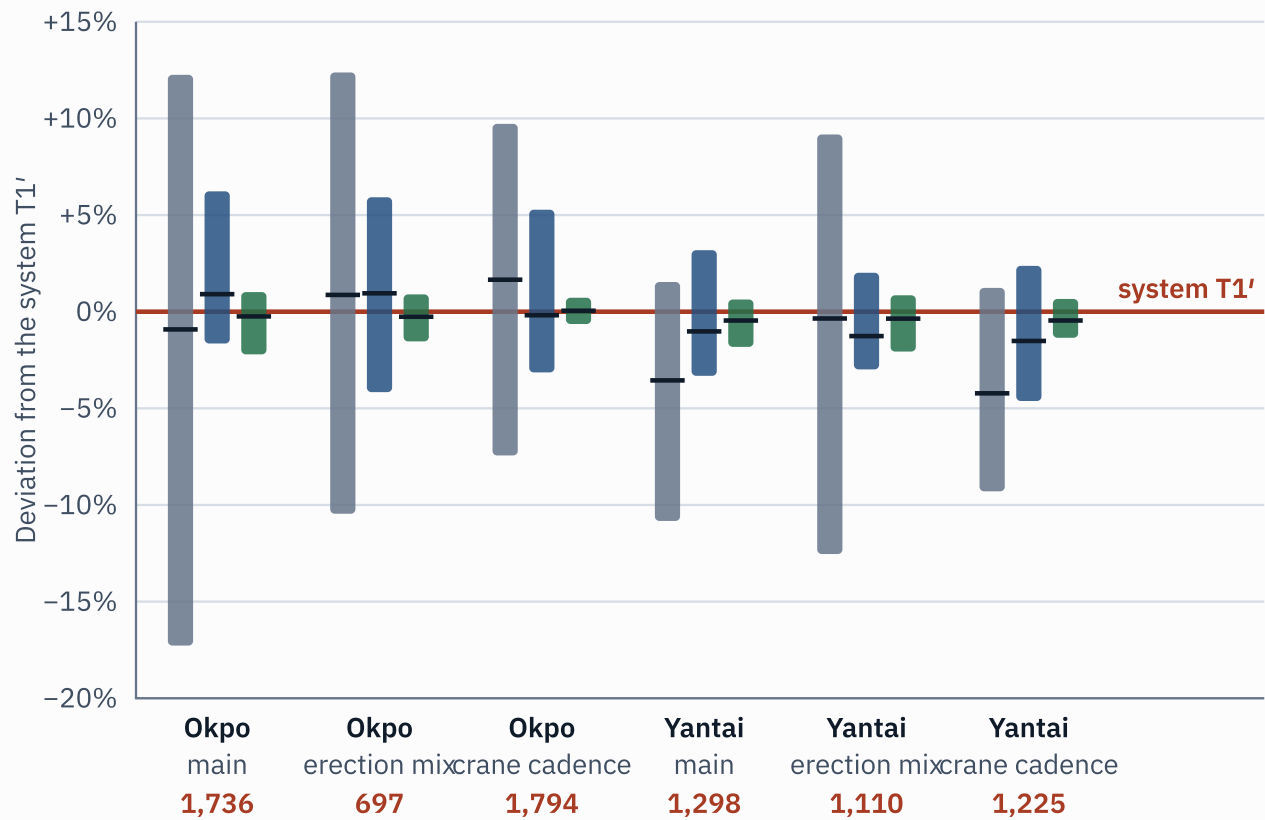
Okpo erection mix (P6 15%), -12%; 10 of 10 seeds bind on road 161; system T1' 697

1,252

Yantai main scenario T1'; 9 of 10 seeds bind at the two doors of building 014; system T1' 1,298. T1 = 1,246 counts the clearance after unloading twice and serves only for comparison

Sources: thesis-core a882742, results/a0ab16fceb94/bounds.csv (Okpo, map rev 1824; T21 confirms rev 1825 has the same network), b82aa19b22af/bounds.csv (Yantai rev 395, same network as rev 396), f7ac88b60111/bounds.csv (erection mix). Conditions: MY-B flow mix, the first 2,000 tasks of each seed's stream; each road one resource. The system T1' (solid line) is from system.csv in results/97d667d16273 and e7f94238b96d (T23). Yantai's binding nodes share one name in the results and were told apart by rerunning the 83c16bf code (00_总控/tasks.md, 9 Oct 2026).

System T1' : expected holds from the sampling law take the seed noise out of chapter 3's bound



SCENARIO	OKPO: SYSTEM T1' · BINDS	YANTAI: SYSTEM T1' · BINDS
main	1,736 road 161	1,298 building 014 door
erection mix	697 road 161	1,110 road 007
crane cadence	1,794 road 010-6	1,225 building 014 door

Proposition draft (Jensen)

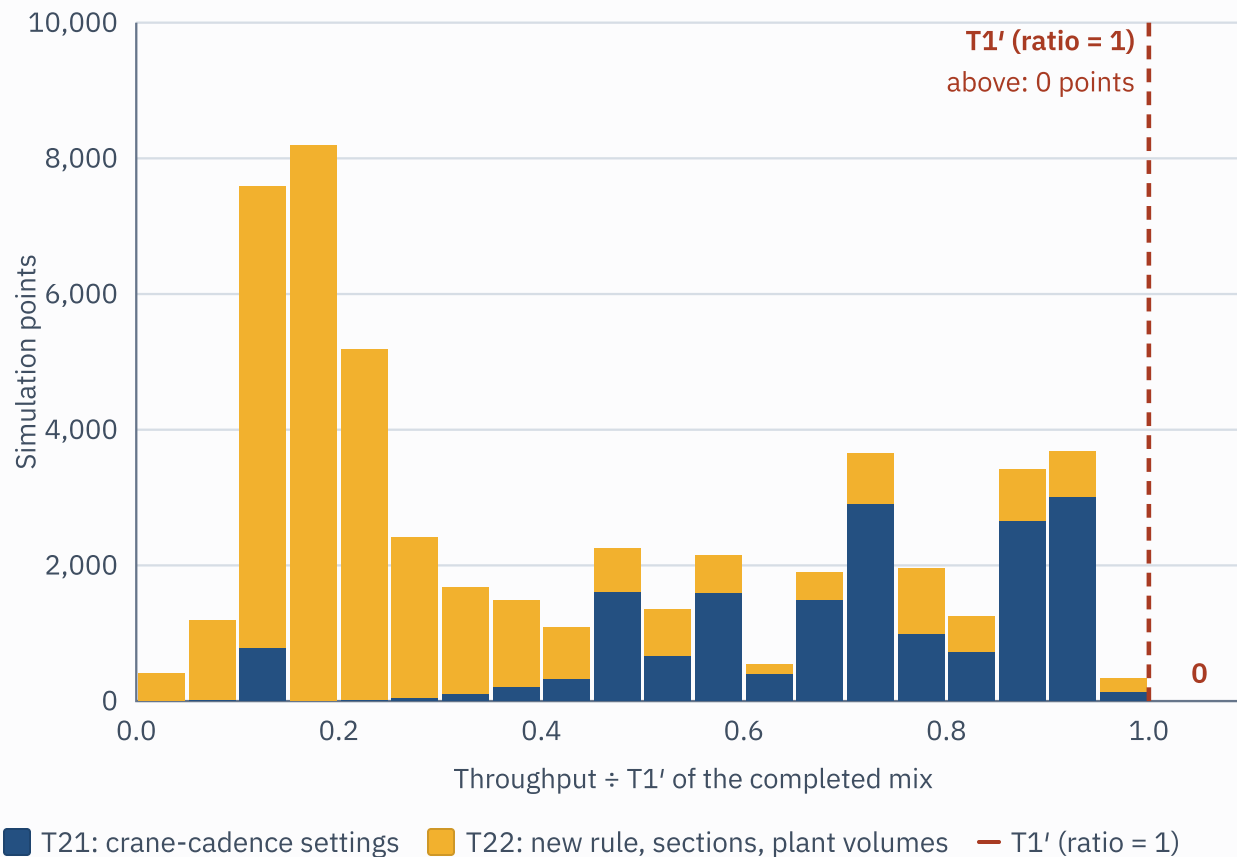
Without parts fixed in absolute amounts, $1/T1'$ is convex in $(\bar{a}, on)$ and sampled mixes are unbiased, so the harmonic mean of sampled $T1'$ is, in expectation, at most the system $T1'$. Of 12 readings, 9 lie below the system value and 3 lie above by less than 1.2 standard errors. The author writes the formal proof.

The same code reproduces the 60 committed per-seed $T1'$ values bit for bit; the system $T1'$ solves in under a second. Chapter 3 quotes the system value; per-seed $T1'$ stays in the case-by-case check.

■ N = 2,000 (per-seed batches) ■ N = 20,000 tasks ■ N = 200,000 tasks — mean of 10 seeds

Sources: thesis-core a882742, docs/T23_report.md §0 item 1 and §1.2–1.4; system.csv and convergence.csv in results/97d667d16273 (Okpo) and e7f94238b96d (Yantai). Conditions: the three scenarios match the committed batches (main, erection mix with P6 at 15%, crane cadence with P6 fixed in absolute amount); each road one resource; maps Okpo rev 1825, Yantai rev 396. Crane cadence has a part fixed in absolute amount and lies outside the proposition; it is shown for comparison.

Checked case by case: none of 51,767 occupancy simulation points from T21 and T22 exceeds T1'



0 / 51,767

Points above T1': of 17,760 in T21 (reservation and the reference rule, peak and saturated) and 34,007 in T22 (adding safe release, the sectioned dock road, plant-level volumes, 16 h and 24 h days)

0.989

Largest ratio: T22, Okpo, $p = 0.99$, sectioned dock road, whole-route reservation, $K = 60$; T21's largest 0.973

4,017 / 4,023

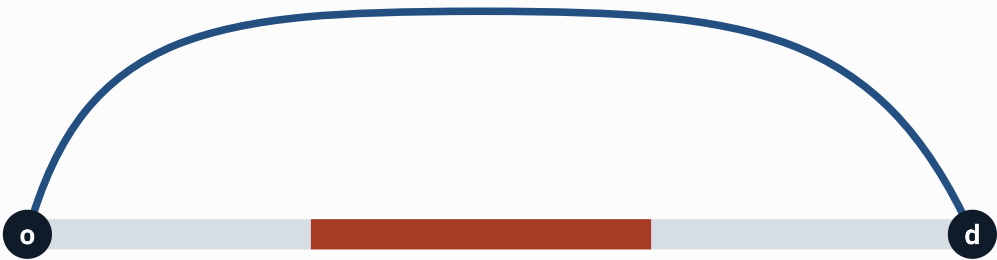
Of the points with ratio ≥ 0.9 , those from settings with nominal crane load $p \geq 0.92$: cranes nearly saturated, bound nearly reached

Before: T18's four batches, 12,140 points, largest 0.949; in T20, 13 points were over by up to 2.3%, traced to the window edge; with the edge term in the code (T21) none is over. T22 also recomputed 410 of T21's points bit for bit; they are not counted twice.

Sources: thesis-core a882742. T21: points.csv in results/1b06c6d46f0d and 87d02d32bc13 (agreeing with T1_tight_check in t21_summary.json). T22: points.csv in results/54f5e26030e2, 16d6dde8b72d, a1f485302418, 656e2da75368, 930e637e0658, ef82da6bf286, dfcdaec9a1e5, c6b4a59a7a82, f1e5a9a4b943, 0fb7e9aefc36 and a79d535a75b3, without free flow and without no-clearing points that complete nothing in the window; sectioned batches use the sectioned network's T1'; consistent with the 18,041 + 22,116 points of the T22 report §8 (5,740 in both rounds). Ratio = throughput ÷ T1' of the mix completed in that point's window (with crane rows and the edge term).

The route-free T1'': loaded moves as multicommodity flow, a bound for any routing rule

T1'': the LP chooses routes (column generation on dual prices)



binding resource (e.g. road 010-6)

T1': loaded moves take the simulation's shortest path

max λ

s.t. $\lambda \cdot F_r + \sum_P y_P \cdot c_{P,r} + \sum_a f_a \cdot \tilde{e}_{ar} \leq H \quad \forall r, k = 0, 1, 2$

$\sum_{P \in P(o,d)} y_P = \lambda \cdot q_{odb} \quad \forall (o, d, b)$

empty-run flow conserved, as in T1'

Counted exactly
route-independent

Loading and tail clearance at the origin, unloading, crane rows, width shares, empty runs (as in T1')

Lower bounds
route-dependent (S1–S7)

Vehicle length at the platform's 16.25 m, no turn sweep, the end stop held only from the last arc, tails cut at the return distance

Certificate
least bound over rounds

μ truncated at zero, v repaired to strict dual feasibility by one super-source Dijkstra: independent of solver tolerance and of convergence

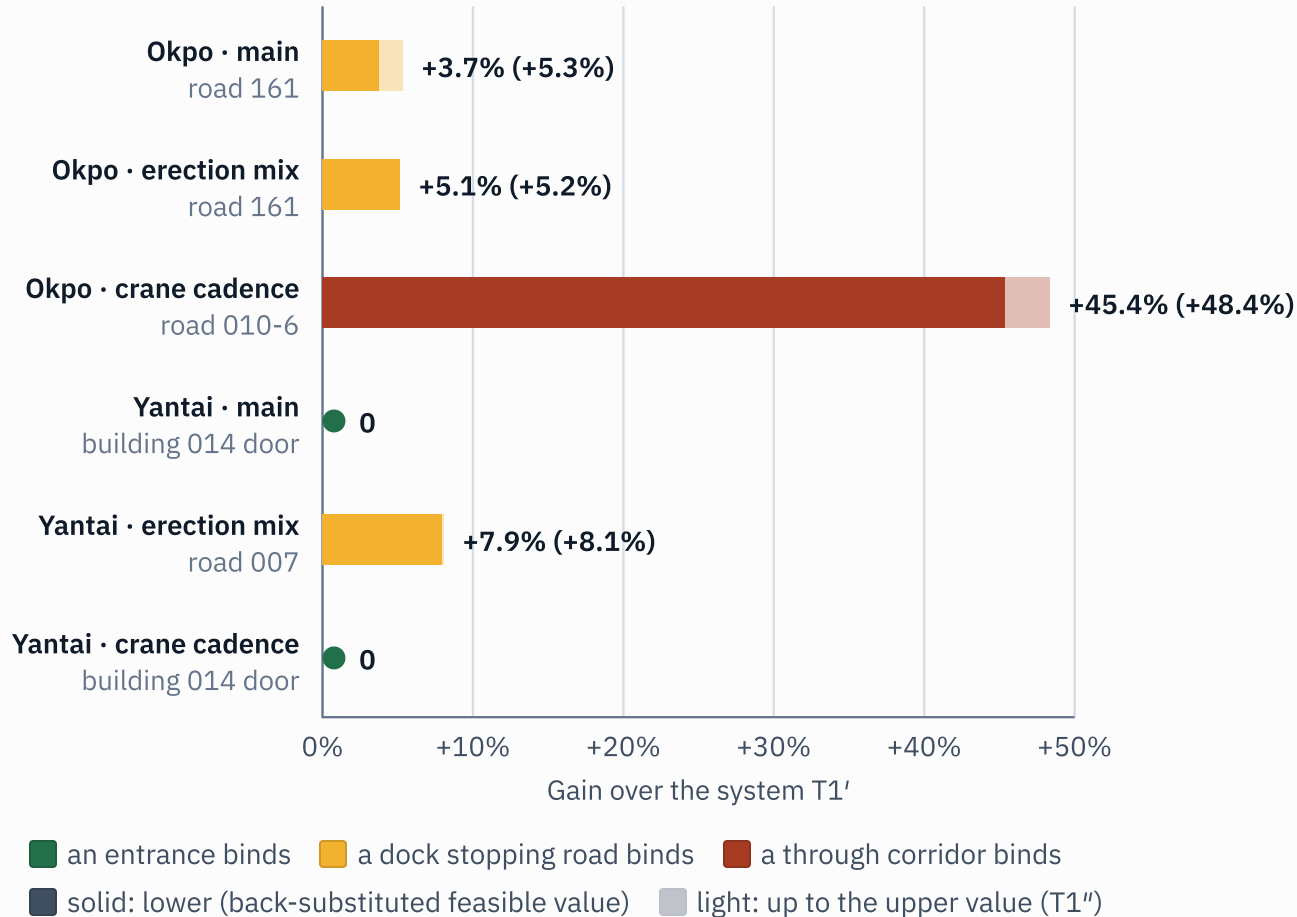
Checks
independent review

$T1' \leq R \leq T1''$ on 60 seeds; no counterexample: a direct arc-flow solve differs by 4.6e-15, 80 random duals, 32,191 detour columns checked resource by resource

Theorem T1'' (suggested wording): for traffic rules satisfying (A1'), (A2), (A3) and any routing rule, $\lambda \leq \min\{T1'', K \cdot H / \bar{c}\}$. The author rewrites the draft proof. The largest LP has 23,316 rows and 29,917 columns; column generation takes 8 rounds, about 6 min.

Sources: thesis-core a882742, docs/T23_route_free_proof.md §1–5 (P'', relaxations S1–S7, (A1'), the Lagrangian certificate); docs/T23_report.md §2.1, §2.5, §5; independent review 00_总控/audit/2026-10-09_T23_review/independent_review.md (c2, c3, c7) and review record 00_总控/reviews/T23.md. R: the value with routes fixed to shortest paths and the same relaxations as T1''. Sketch not to scale.

Route slack in the bound as an interval: zero at an entrance, +4–8% on a dock stopping road, +45% on a through corridor



Entrance

Yantai main, crane

Zero by construction: the binding entrance has a single incoming arc, so its holds come from handling and the one last stretch, the same on any route

Dock stopping road

Okpo 161, Yantai 007

Small through share (0.3–0.7%; 20%). The slack comes mainly from local flow entering the stops from the other end, plus a little through traffic going round

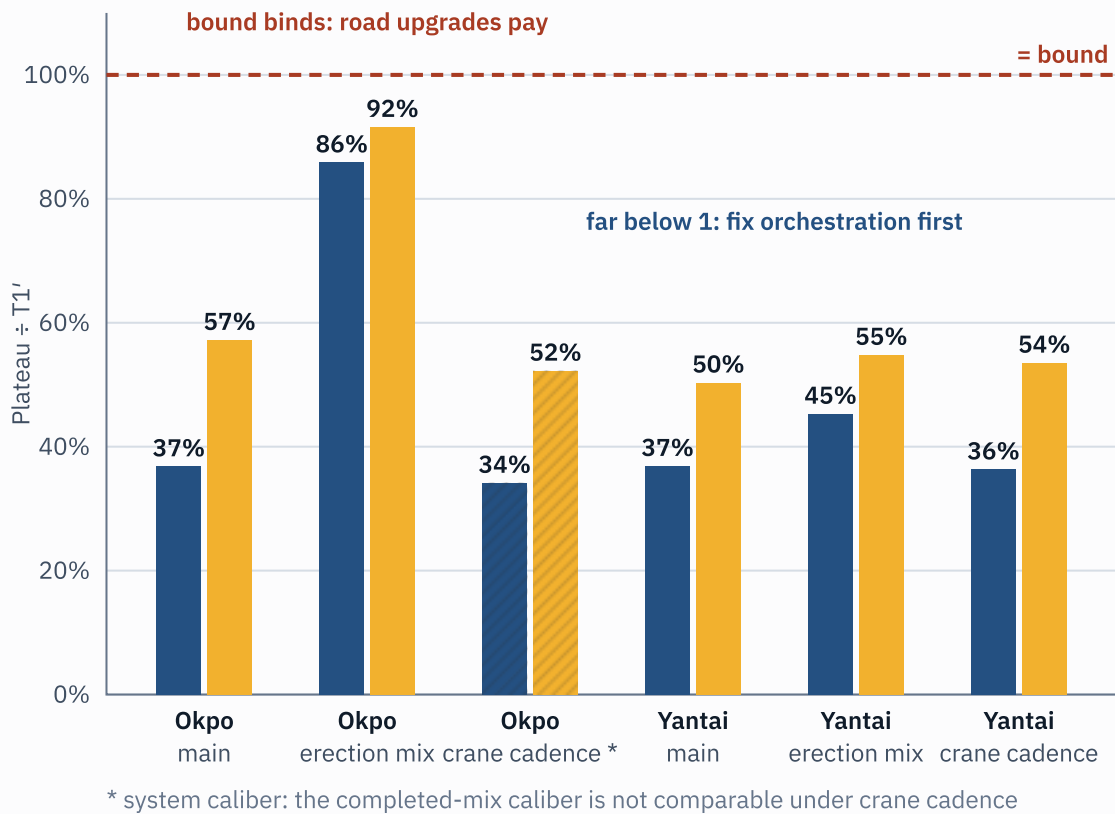
Through corridor

Okpo crane, road 010-6

97% of its 462 loaded moves a day pass through; the detour around it still has 22% to spare and adds 235 m on average

Lower = back-substituted feasible value – system T1': keep T1'''s route split, restore every coefficient to the unrelaxed T1' form and re-optimize empty runs; upper = T1'' – system T1'. It caps what detour orchestration could recover, and chapter 4 measures how much it does; each class has only 1–3 points, and T25 adds more before the wording becomes a mechanism.

Two uses of the bound: plateau over bound separates C3 from C4, and dual prices in two columns give the exchange rate



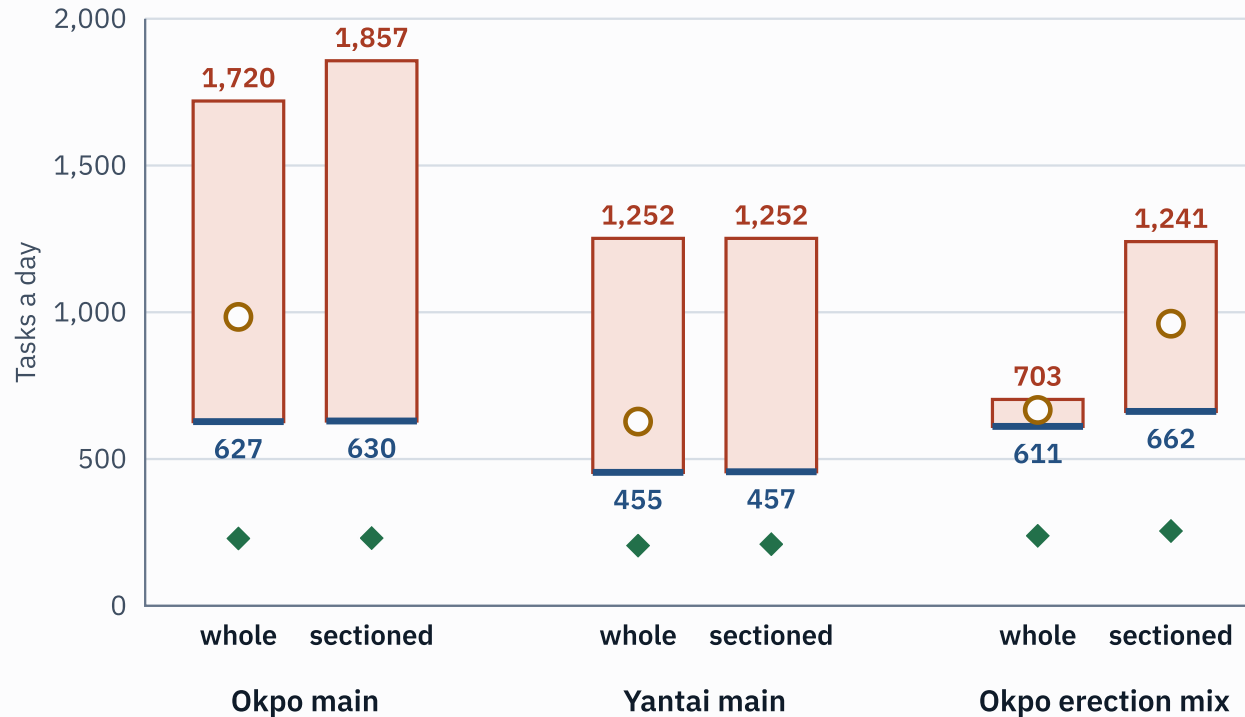
■ whole-route reservation ■ reference rule (teleport clearing)

RESOURCE · SCENARIO	SHORTEST-PATH DISPATCH (T1')	DETOURS ALLOWED (T1'')
road 161 Okpo · main	holds -10%: +134 (first-order +193) widen 1 m: 0	holds -10%: +203 widen 1 m: 0
road 010-6 Okpo · crane cadence	holds -10%: +198 widen 1 m: +188	0
road 013-7 Okpo · crane cadence	0	holds -10%: +283 (first-order +295) widen 1 m: +164
building 014 door Yantai · main	holds -10%: +2.3 (first-order +144)	same (entrances do not depend on routes)
road 007 Yantai · erection mix	holds -10%: +123 widen 1 m: +127	holds -10%: +133 widen 1 m: +138

Tasks a day. “Holds -10%” means 1.78 extra hours a day; re-solved values, with the first-order estimate in brackets where it differs. Where the two columns disagree, road upgrades and orchestration can stand in for each other; width pays in steps, so T25 prices “widen to the next class”.

Sources: thesis-core a882742, docs/T23_report.md §4.1–4.4; platform.csv and prices.csv in results/97d667d16273 and e7f94238b96d. Plateau = mean saturated throughput over K = 100–150 in the committed batches (shortest-path routing); divided by the T1' of the batch's completed mix, or by the system T1' for Okpo's crane cadence (the completed-mix caliber scales P6 up and reads as crane utilisation). Yantai's two building-014 doors are both full: upgrading one barely helps, both together give +144 at first order.

Sectioning the dock road: in the main scenario only the upper edge rises (+8%); where the dock binds both edges move (+76.5% / +8.3%)



- interval: reservation plateau → T1' ◆ safe release (segment_safe) plateau
- reference rule (teleport clearing) plateau

- **Okpo main:** T1' 1,720 → 1,857 (+8%), binding moves from roads 161 and 010-6 in turn to 010-6; reservation plateau 627 → 630 (+0.3%): the rule, not this road, limits throughput
- **Okpo erection mix:** T1' 703 → 1,241 (+76.5%), reservation 611 → 662 (+8.3%); plateau over T1' 0.87 → 0.53: change the granularity and the bottleneck moves from the road to the rule
- **Yantai:** an entrance binds, so sectioning leaves T1' at 1,252
- **Still a bound after sectioning:** resources are only finer, (A1)–(A3) still hold and the proof is unchanged; on one mix the sectioned T1' is never below the whole-road one

The lower edge is the better of the deadlock-free rules tested so far: whole-route reservation. Safe release plateaus at 205–255, with its peak at $K \approx 18$ –31, below reservation throughput. The widening to 12 m (1,805 / 786) and the passing bays of the author's probe get ten-seed numbers in T25.

Sources: thesis-core a882742, docs/T22_report.md §2.3, §2.6, §3.5, §4; results/t22_report_3287d62c7916/t22_interval.csv and the batches' points.csv and summary.json (main: Okpo 656e2da75368, Yantai 930e637e0658; erection mix a1f485302418; whole-road reservation and the reference rule reused from 49d2383872e7, b82aa19b22af, dfb2758822ad). Conditions: MY-B flow mix, mean over saturated $K = 100$ –150, 10 seeds, 4 days; sectioning = block boundaries 25 m either side of every stop on roads longer than 200 m with roadside stops (exactly the 4 dock roads of the two yards), a scenario setting; the reference rule has direction locks when sectioned; maps Okpo rev 1825, Yantai rev 396. The widening probe is in appendix A of the author's review (9 Oct 2026).

Both yards' binding elements sat on conservative modelling choices: the key-element review becomes a fixed step

Okpo · road 161 (9 m wide, 1,095 m, at dock no. 1)

Main scenario: binds in 5 of 10 seeds, dual price 98.9 a day per extra hour (about 4.2 vehicles); road 010-6 in 4, 115.5. Erection mix: 10 of 10, 43.9 (1.8 vehicles).

Author's imagery check: with a vehicle unloading or waiting for the crane at the dock edge, the neighbouring assembly areas stay reachable. → From T22 Okpo's main scenario is sectioned at the stops (T1' 1,857).

Yantai · the two doors of building 014

Binds in 9 of 10 seeds. Both doors are dead ends where vehicles load and unload, holding the door about 10 min; with only two doors, each serves about 7% of all tasks.

Author's imagery check: two doors with room to pass in front, so exclusive loading at the door is conservative. T23: both doors full (1.000 / 0.998); one door adds 2.3 a day, both +144 at first order. → T25: a door-granularity scenario, the two doors as one upgrade.

The fixed step (every yard in chapter 6)

- 1 Compute the system T1' and every resource's dual price (shortest-path dispatch and detours allowed)
- 2 List the three elements with the highest dual prices
- 3 The author checks on imagery: **road width, number of entrances, how traffic passes a stop**
- 4 Choose the resource granularity, recompute, and record it in the map's source field

This is a pinpoint check of a few key elements, not a precise survey; it can go into the thesis as an engineering method (minimal human check).

What stands and what comes next: every computation lands in a figure, a table or a section of the thesis

In hand (T17–T24)

- ✓ The T1' LP with its three kinds of rows, code, tests and a proof draft
- ✓ The system T1' and the route-free T1'' with its certificate; route slack as an interval (T23)
- ✓ 51,767 occupancy points checked, none over (T21, T22)
- ✓ Safe release exact and never deadlocked; proof unchanged with the dock road sectioned (T22)
- ✓ Citations checked; the nearest same-field work told apart in full text (T24, T24b)

Conditions (stated once): system T1' and T1'' bound the steady state, the case check uses each point's completed mix; T1' fixes loaded routes to shortest paths, T1'' leaves them free; the reference rule (teleport clearing) is outside the proof; T23 treats each road whole, Okpo's main scenario is sectioned from T22 (1,857); 16-h working day; MY-B flow mix; maps Okpo rev 1825, Yantai rev 396, and from T27's step zero the maps in 20_数据/船厂地图 (Okpo rev 1833), kept by T25.

TASK	WHAT	OUTPUT (PLACE IN THE THESIS)	STATUS
T27	A less conservative deadlock-free rule: Banker plus a step-wise feasibility search, or claims cut at the next safe holding point (T5); re-checks the lower edge. Step zero: Okpo map rev 1833	Rule comparison (ch. 4); lower edge of the interval table (ch. 3)	running
T25	Upgrade simulations and the first exchange-rate table; door granularity; capacity against block length; the three-resource dock network	Rate table (ch. 5); Yantai's T1' and binding elements; the dock section	after T27
T26	Floating docks; a finite holding area S	H7 case; holding areas (ch. 5)	after T25
Author	Rewrite the T1' and T1'' proofs; settle the wording of the Jensen proposition and the three readings	Chapter 3, theorems and method	weeks 3–4
–	Candidate: attainability of T1' (a periodic rule reaching $1 - \epsilon$ of T1' when the dock dominates)	Chapter 3, theorems	to do
–	Complexity: NP-hard (from the blocking job shop)	Chapter 3, theorems	to do

Sources: rows T22–T27 of 00_总控/tasks.md (10 Oct 2026; the author set T27 before T25 that day); review records 00_总控/reviews/T22.md, T23.md, T24.md; the T27 task statement 00_总控/runs/2026-10-10_T27_prompt.md; manifesto v1 §4. Running and queued tasks are plans, not results.

Risks and mitigations: each has a way to strengthen it and a place in the thesis; the plan stands

RISK	MITIGATION (HOW TO STRENGTHEN IT)	OUTPUT
T1" takes the vehicle length at its lower bound; on detours this matters about as much as R – T1'	Back-substitution already gives the lower value; split commodities into 3 length classes (mean shortfall 5.2 m → 1.5 m), warm-started from the unsplit column pool	Narrower route-slack interval (ch. 3); an appendix table
Degenerate optima put the price on a single resource	Re-solves already give exact gains; T25 re-solves combined upgrades (both building-014 doors) and runs a relax-and-find-the-next-bottleneck sequence	Combined-upgrade rows of the rate table (ch. 5)
Safe release is conservative (about 2/3 of waits start from refusals; the ring6 case), so the lower edge is reservation for now	T27: step-wise feasibility search or claims cut at the next safe holding point, reporting the refusal share and feeding the same batches into the interval table	Rule comparison (ch. 4); the lower edge
The 200 m and 25 m of sectioning are scenario settings	The author checks road 161 and Yantai's roads 007 and 014 on imagery; T25's block-length scan turns the 25 m into a curve	Scope line of the dock section; “control granularity” (ch. 5)
Only 1–3 points per binding-resource class	Every T25 upgrade scenario reports the through share, the alternative route's slack and the route-slack interval, drawn as a scatter	“Upgrade the road or the orchestration” (ch. 5)

Sources: thesis-core a882742, docs/T23_report.md §7 (risks and how to strengthen) and §2.6; docs/T22_report.md §2.1 and §10 items 1 and 2; review records 00_总控/reviews/T22.md and T23.md. None changes the plan.

What is new: route-fixed and route-free LP bounds for networks with shareable widths and asymmetric occupancy

- **Within our search** (the author's review, appendix B, 9 Oct 2026), the shipyard transport literature has no capacity bound; the nearest same-field work, checked in full text (right), does not narrow the claim, and the introduction names it
- We extend the occupancy-rate LP to road networks with **shareable widths and asymmetric loaded and empty occupancy**, valid for any rule satisfying (A1)–(A3)
- Dual-feasible functions serve as **time-integrated capacity rows**; crane rows and the window edge term bring the dock into the same bound
- **A route-fixed bound (T1')** beside a route-free one (T1''), separating the route slack and its sources; whether the field already has a route-free bound under occupancy is still to be searched
- **As a system**: tracing on public imagery → bound → dual prices (two columns) → key-element review, one LP per yard

Nearest same-field work: Tao et al. 2017 (full text)

ITEM	TAO ET AL. 2017	THIS STUDY
Road capacity	in vehicles: 2 on main roads, 1 on side roads	width containers and blocking times that change with load
Traffic rules	one: first come first through, wait or reroute	four rules side by side, incl. deadlock-free release
Deadlock	not handled	reservation and safe release, deadlock-free by construction or by test
Capacity bound	none, simulation only	T1', system T1', T1'', checked case by case
Yards	one, tuning vehicle types by load	two, traced on public imagery

Sources: the author's review §1.1 and appendix B; the library notes 00..小论文/Reference/notes/occupancy_anchors_20261009.md §3.1 (Tao et al. 2017, card 03-055, full text, T24b: item-by-item comparison from §2.1, §2.3, §2.4, §3 and figure 5); T23 report §6.

Sources we build on: same-mechanism work from other fields supports and informs the study; it is not prior coverage

SOURCE WE BUILD ON	WHAT WE TAKE	READ
Railway blocking times, UIC 406; Goverde 2007	Blocking time → occupancy rows; max-plus gives the cycle of a given order, a lower-side result	abstract; UIC 406 to check
Jensen, Schmidt, Nielsen 2020	Railway network capacity without a fixed timetable, computed by branch and bound: structurally among the closest; “ignoring junction conflicts overstates capacity” matches junctions as exclusive resources	full text
Campos, Chiola, Silva 1991	LP throughput bounds for Petri nets whose conflict decisions keep relative frequencies, the same structure as T1’s fixed routes; T1” frees the routes and goes beyond that class	full text
Queueing networks with blocking: Onvural 1990; Balsamo et al. 2001	Theory of blocking and starvation	abstract; book to check
Fekete & Schepers 2001	Dual-feasible functions (bin-packing lower bounds) → width bin-packing rows	lemma proved in the draft
Banker's algorithm (Dijkstra); Habermann 1969	The safe-state test → safe release, segment_safe	to check
Mascis & Pacciarelli 2002	Blocking job shop → complexity reduction	abstract

Sources: the author's review, appendix B (searches and reading depth); the library notes occupancy_anchors_20261009.md §4.3 (Campos et al. 04-094, Jensen et al. 04-095, full text); T22 report §13 (Banker's algorithm to check); T18 proof draft §6.2 (max-plus is on the lower side). The phrasing is “we extend X to ...” and “building on X ...”.

Target journals for paper 1 (chapters 2–3): class B first, class A as a stretch

TIER	JOURNAL (SCHOOL CLASS)	WHY IT FITS / CONDITION
First	Transportmetrica A: Transport Science (class B)	Transport science at the core: the occupancy model, a rule-free bound and the capacity interval, with rigorous modelling and strong baselines
First	Ocean Engineering (class B)	Shipyards transport mechanisms and resource limits as the main line: two yards, tracing on public imagery, the dock and erection
Stretch	Reliability Engineering & System Safety (class A)	Condition: the dock-mouth proposition is written and proved as a stability or resilience result, “blocking-induced starvation, instability and recovery”
Alternative	IEEE Transactions on Automation Science and Engineering (class B)	Capacity and release rules of automated logistics systems (safe release, reservation)
Alternative	Applied Mathematical Modelling (class B)	The model and bounds themselves: width bin-packing rows, system T1', T1'' and its certificate
Supplementary	Computers & Industrial Engineering; Journal of Ship Production and Design (class C)	Well-matched supplementary outputs

Class B first; class A as a stretch, with its condition in the table. Graduation needs at least one paper in a class-B-or-above journal (MY-A is already published in Ocean Engineering, class B).

Sources: the School of Shipbuilding Engineering 2026 journal directory (graduation requirements file, version of 23 Sep 2026), §1, §2, §7; the paper plan of 10 Oct 2026 (paper 1 covers the bound, capacity and dock sub-studies). “Why it fits” is our own matching of this sub-study to each journal, to be checked against each journal's aims and scope before submission.

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Data and documents: thesis-core commit a882742 (T22, T23 reviewed 10 Oct 2026; T17–T21 files as at 83c16bf): result directories (in each slide's source line), reports and proof drafts T17–T23; manifesto v1; the author's review of 9 Oct 2026; review records 00_总控/reviews/T22–T24.md; library notes occupancy_anchors_20261009.md; 00_总控/tasks.md; the school's 2026 journal directory; the DCY270 data sheet. Charts: bound/make_data.py.

What this sub-study gives the thesis: a network bound that can be computed, checked and that points the way

1

Computable

One LP per yard: the system T1' in under a second, T1'' by column generation in at most about 6 min; only networks traced on public imagery and public parameters.

2

Checked

Proof drafts for T1' and T1'', with a certificate for T1''; none of 51,767 occupancy simulation points exceeds the bound; the independent review found no counterexample to T1''.

3

Points the way

Dual prices in two columns name the binding roads and doors, and show where road upgrades and orchestration can stand in for each other.

← Back to the thesis briefing

Capacity interval →

Dock and erection peak →

Fleet-network co-design (exchange rate) →

Transporters own the road: capacity is capped by the network, and the bound tells us how high the cap is and where it binds.

Thank you.